EXPERIMENT 8 • LAB RECORD

Vehicle Service Request Management System

Configuring user management and role-based access control (RBAC) in a service-request application

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Register number	241401110
Application	Vehicle_Service_Request_Management_System
Platform	Zoho Creator (low-code application platform)

1. Aim

To configure user management and role-based access control for the Vehicle Service Request Management System in Zoho Creator. The experiment focuses on reviewing permissions, arranging a role hierarchy, and defining a data-sharing rule so that the Service Manager can read and update service requests submitted by the Service Advisor.

Objectives

• Review the service-request dashboard and the service request report.

• Study the available permission sets for reading and updating application data.

• Create a role hierarchy: System Administrator → Service Manager → {Service Advisor, Technician}.

• Configure a share rule named **ServiceAdvisor_to_ServiceManager** with Read/Write access to service-request records.

• Distinguish record ownership from record visibility and apply the principle of least privilege.

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Software and tools

• Zoho Creator and the Vehicle_Service_Request_Management_System application

• Web browser with internet access

• Zoho account with permission to edit the application and configure user permissions

2. Theory

User management and role-based access control

User management determines who can access an application and what actions they may perform. Role-based access control (RBAC) assigns permissions to job roles rather than separately configuring every user. In Zoho Creator, permission sets, roles, and data-sharing rules work together to control access to forms, reports, dashboards, and records.

Concept	Meaning
Users	People added to the application who sign in to submit, review, or manage service requests.
Permissions	Sets of allowed actions, such as viewing records or creating and updating information.
Roles	Organizational positions that determine a user's place in the access hierarchy.
Role hierarchy	A reporting structure that can provide higher-level roles visibility into records owned by subordinate roles.
Data sharing	A rule that grants specified roles additional access to records at Read Only or Read/Write level.

Application specification

Parameter	Detail
Application	Vehicle_Service_Request_Management_System
Dashboard / report	Vehicle Service Dashboard and Service_Request_Report
Permission sets	Read and Write, according to the actions required
Role hierarchy	System Administrator → Service Manager → Service Advisor and Technician
Custom share rule	ServiceAdvisor_to_ServiceManager: Service Advisor → Service Manager, Read/Write

241401110

3. Procedure

1. Sign in and open the application. Sign in to Zoho Creator and open the Vehicle_Service_Request_Management_System application from the application home page.

2. Review the dashboard and report. Open the application in Access mode. Review the dashboard and the service request report to understand how request details, vehicle information, customer contact details, assigned technician, and service status are presented.

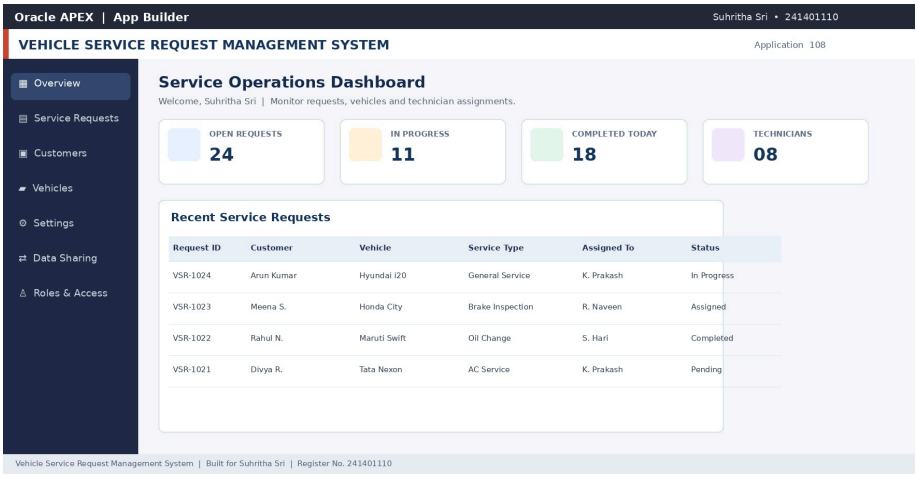


Figure 8.1: Vehicle service dashboard under Suhritha Sri account.

3. Enter Edit mode and open Settings. Choose the option to edit the application, then open Settings from the application navigation.

4. Open User Permissions. Under the Permissions area, open User Permissions. Review the available Permissions, Roles, and Data Sharing sections. Identify which permission sets are needed for users who submit requests and for staff who manage them.

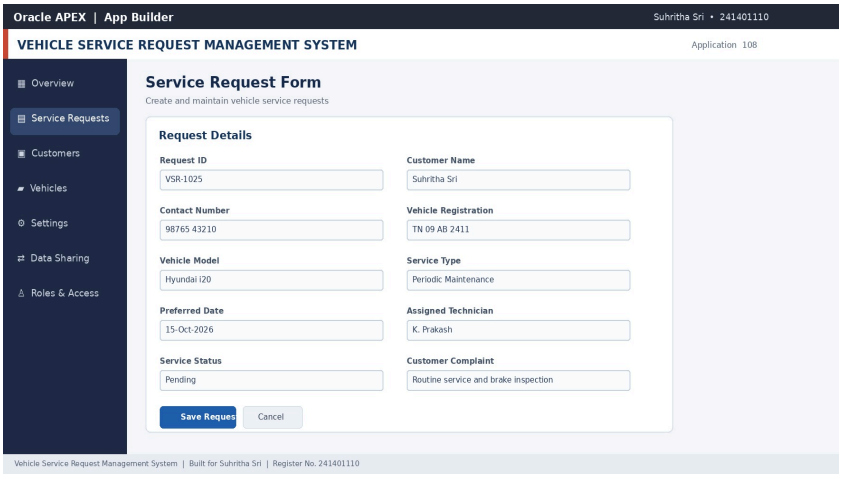


Figure 8.2: Service request form with customer and vehicle details.

5. Configure the role hierarchy. In the Roles section, arrange the roles as shown below. The System Administrator has overall oversight; the Service Manager supervises the service workflow; the Service Advisor records customer requests and the Technician handles assigned service work.

Level	Role	Intended responsibility
1	System Administrator	Application configuration and overall access oversight.
2	Service Manager	Reviews incoming requests, assigns work, and updates service status.
3	Service Advisor	Captures customer, vehicle, complaint, and booking details.

Level	Role	Intended responsibility
3	Technician	Reviews assigned jobs and records inspection or repair progress.

241401110

3. Procedure (continued)

6. Create the data-sharing rule. Open Data Sharing and select New Share Rule. Enter the rule name **ServiceAdvisor_to_ServiceManager**. Select the service request form, set the source role to Service Advisor, set the receiving role to Service Manager, and choose Read/Write access.

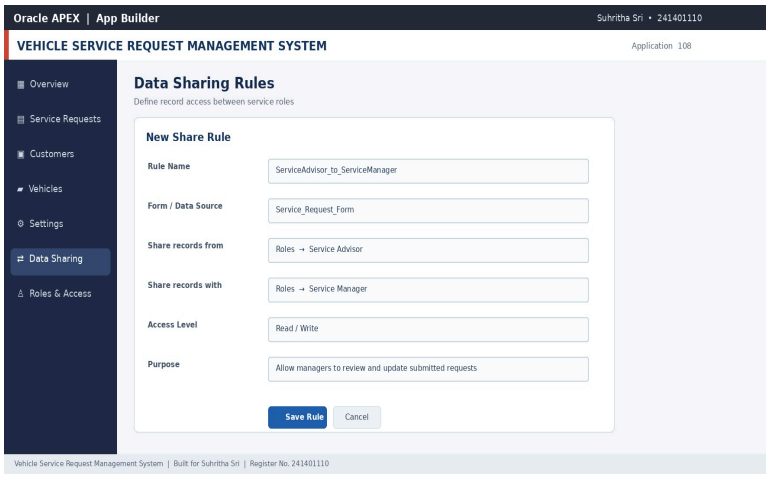


Figure 8.3: Data-sharing rule for Service Advisor and Service Manager.

7. Save and verify. Save the rule and confirm it appears in the Data Sharing list. Reopen the saved rule and check its name, form, source role, receiving role, and access level. If any value is incorrect, edit the rule and save it again.

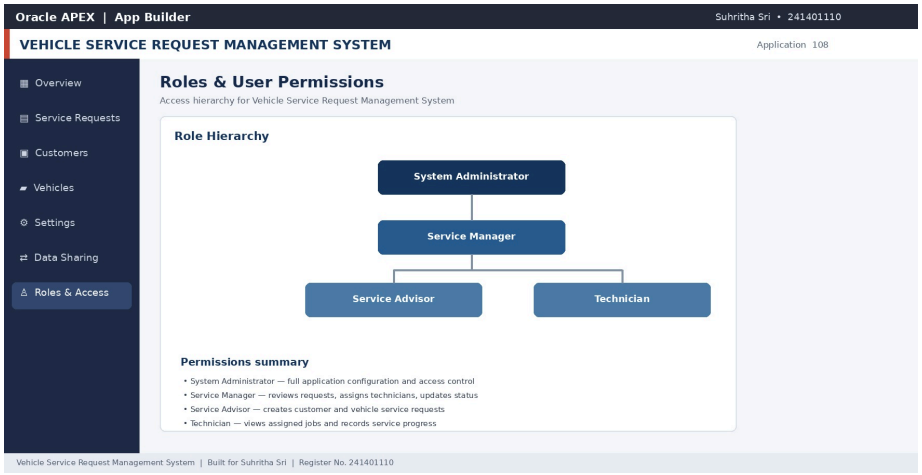


Figure 8.4: Role hierarchy and access permissions for service staff.

Expected configuration

Setting	Configured value
Rule name	ServiceAdvisor_to_ServiceManager
Form	Service_Request_Form
Share records from	Roles → Service Advisor
Share records with	Roles → Service Manager
Access level	Read/Write

4. Role hierarchy and data sharing

The role hierarchy establishes the normal access structure, while a specific share rule extends access where the workflow requires it. The Service Advisor owns the submitted request records. The Service Manager receives Read/Write access through the named rule, enabling review and status updates without granting unrestricted write

access to every user.

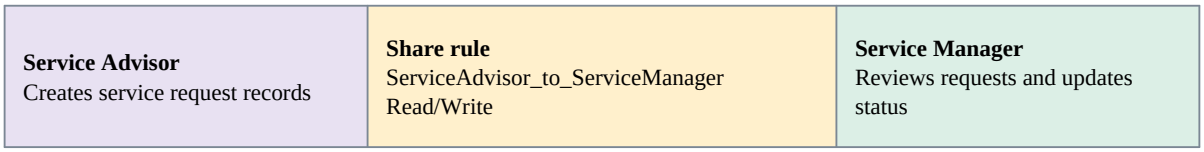


Figure 8.5: Flow of service-request record access through the share rule.

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RBAC working summary

Level	Role	Access purpose
1	System Administrator	Maintains the application and oversees access configuration.
2	Service Manager	Monitors requests, assigns technicians, and updates status for shared records.
3	Service Advisor	Creates and maintains customer and vehicle service request details.
3	Technician	Works on assigned requests and records inspection/repair progress according to granted permissions.

Verification checklist

Check	Expected result
Application dashboard reviewed	Service request report and key request fields are identifiable.
Permission sets reviewed	Read/Write capabilities are understood and applied as needed.
Role hierarchy configured	System Administrator, Service Manager, Service Advisor, and Technician are arranged correctly.
Share rule saved	ServiceAdvisor_to_ServiceManager is listed under Data Sharing.
Access verified	The Service Manager can read and update the intended shared records.

5. Result

The user-management and RBAC configuration for the Vehicle Service Request Management System was specified. The role hierarchy and the ServiceAdvisor_to_ServiceManager Read/Write sharing rule provide a controlled way for service requests to be entered by the Service Advisor and reviewed or updated by the Service Manager. The configuration should be verified in the live Zoho Creator application before it is treated as implemented.

6. Conclusion

This experiment demonstrates how permissions, roles, a role hierarchy, and data-sharing rules can support secure service-request management in Zoho Creator. Separating responsibilities between the System Administrator, Service Manager, Service Advisor, and Technician helps keep access aligned with job duties. A targeted sharing rule supports coordination while following the principle of least privilege.

Note: Screenshots retained from the provided reference PDF are illustrative reference images; they may display the original appointment-booking application rather than the vehicle-service application. Replace them with screenshots from the configured vehicle-service application if required for final submission.

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